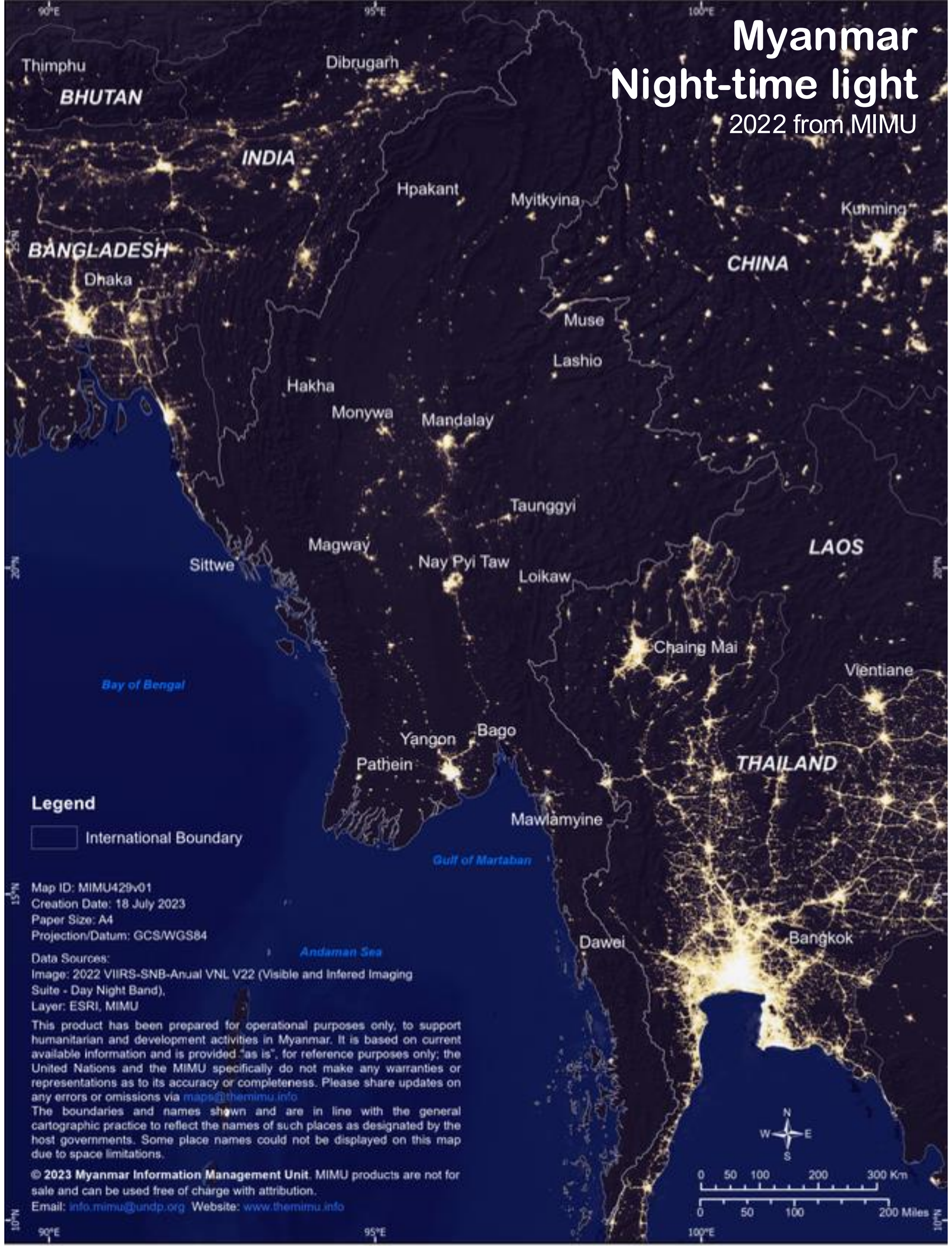


Geospatial Analysis of Illicit Economies in Post-2021 Myanmar coup d'état Using Nighttime Lights

Jirawat Poongam (MEP23207)
Advisor: Assistant Professor Jun Goto

Motivation

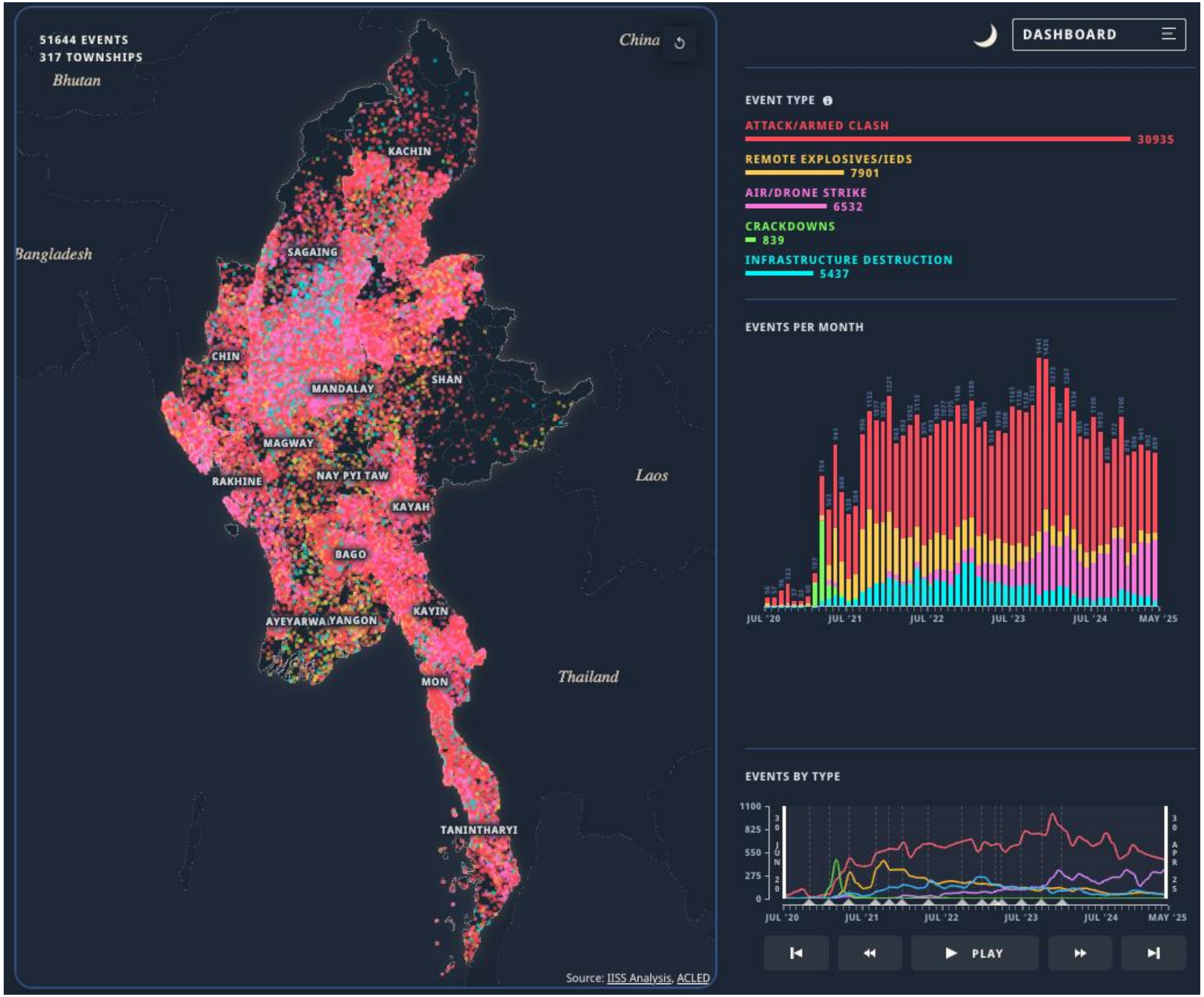


Figure 1: Myanmar conflict map
2020-2025
from ACLED created by IISS

Humanitarian Impact:

The conflict has resulted in significant civilian casualties, displacement, and destruction of properties.

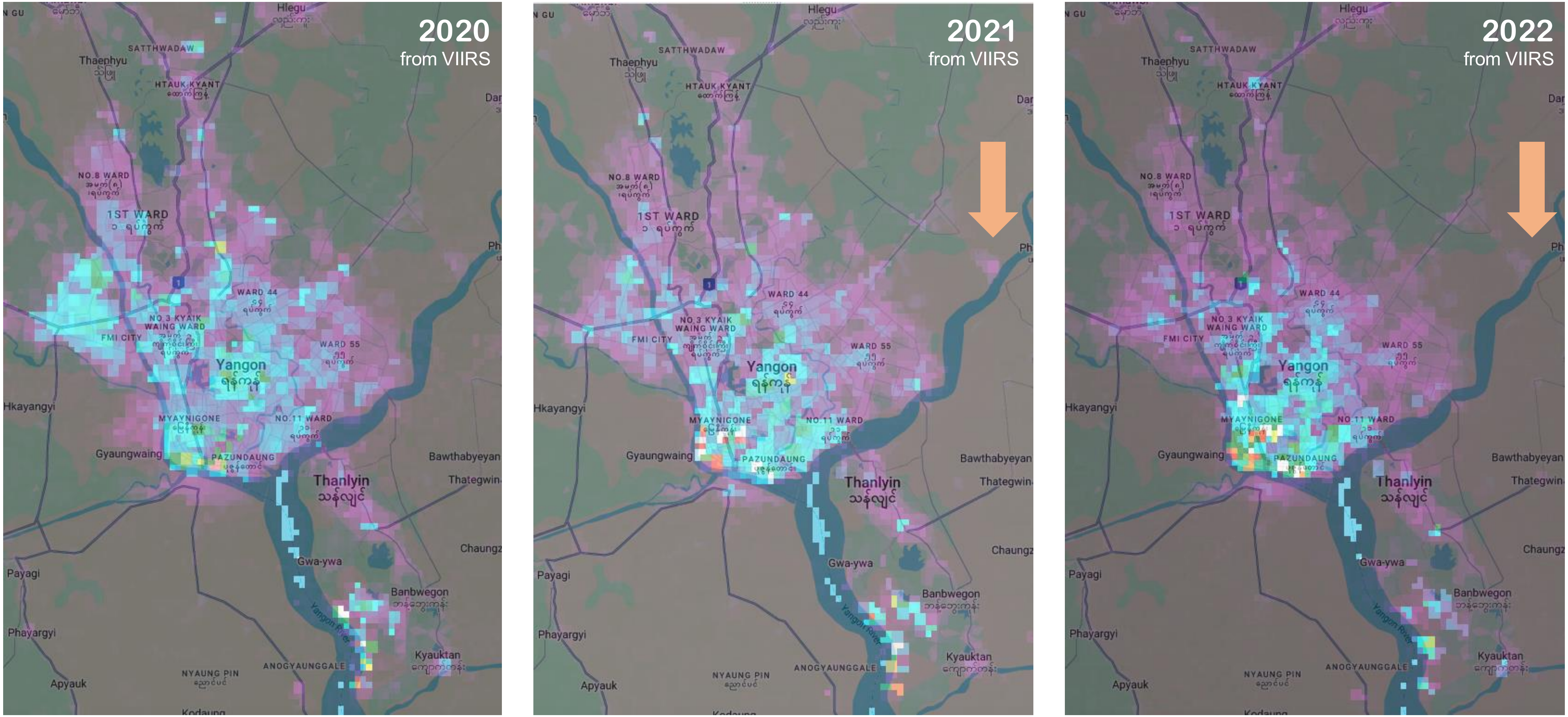
Economic Impact:

The war has severely affected Myanmar's economy, including disruptions to trade and economic activities

Motivation

Observed activities from space using night-time light

Figure 2: Average night-time light from in 2020, 2021, and 2022 from VIIRS calculated by NASA

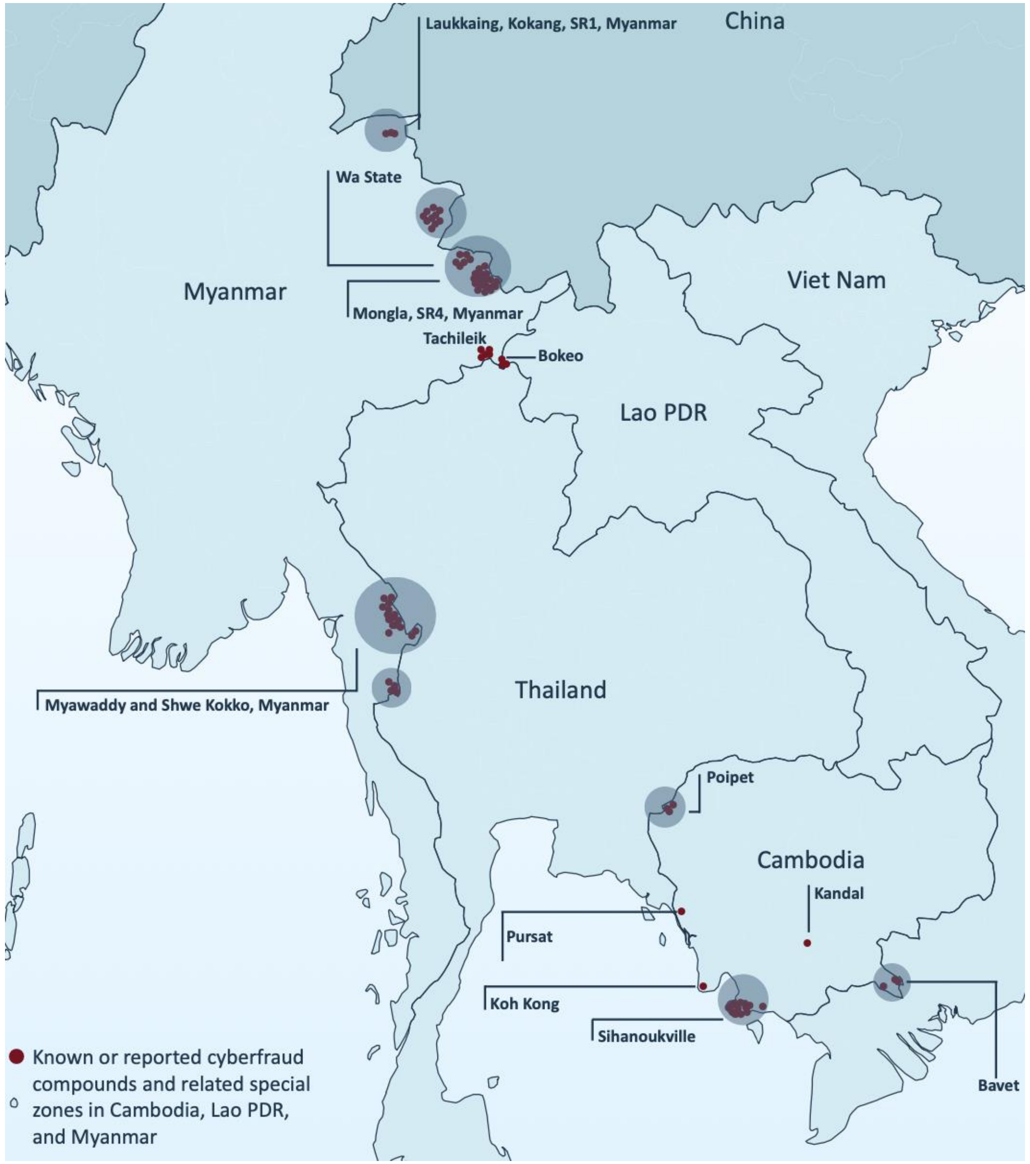


Motivation

Transnational organized crime (TOC)

- **Drug Trafficking**
  The illegal production, transportation, and distribution of drugs.
- **Human Trafficking**
  The exploitation of individuals for sexual or labor purposes.
- **Money Laundering**
  The process of disguising the origins of illegally obtained money.
- **Cybercrime**
  Criminal activities carried out using computers and networks.
- **Smuggling**
  The illegal movement of goods (including migrants) across borders.
- **Illicit Trading in Firearms**
  The illegal trade of weapons and ammunition.

Figure 3: Locations of known or reported cyberfraud compounds and related special zones in Cambodia, Lao PDR, and Myanmar, 2023

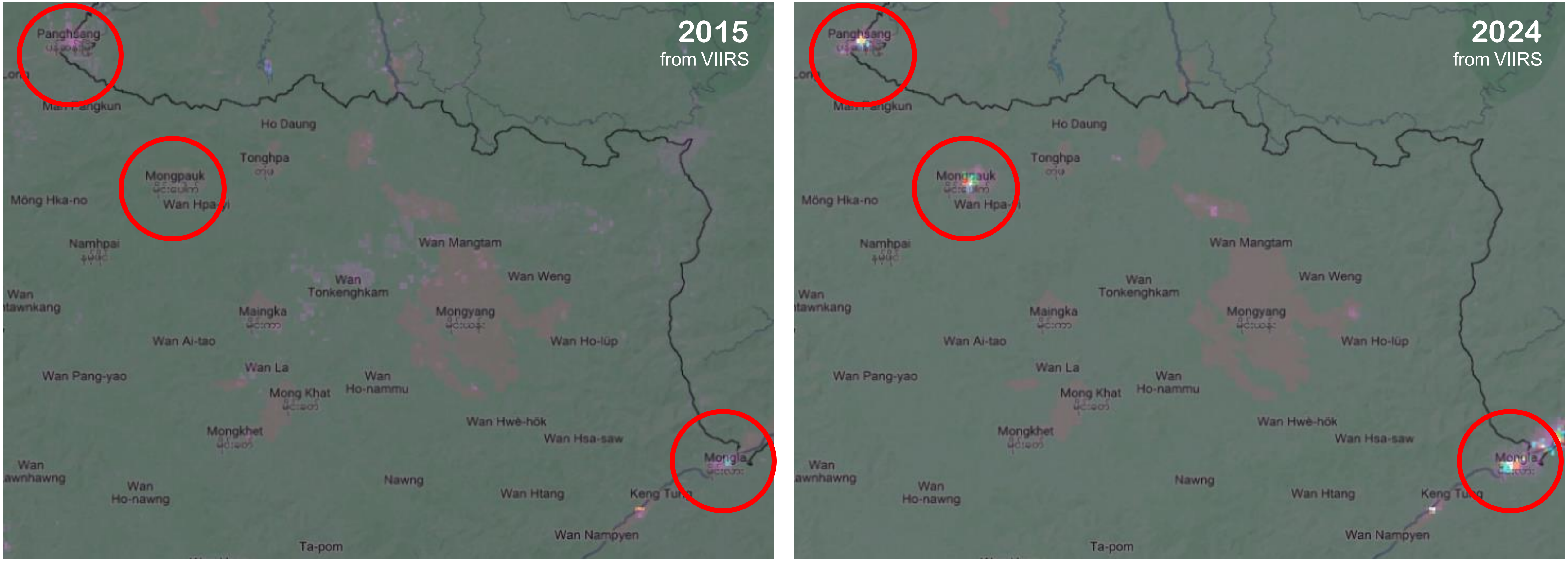


Source: UNODC. (2024). Casinos, Money Laundering, Underground Banking, and Transnational Organized Crime in East and Southeast Asia: A Hidden and Accelerating Threat

Motivation

Observed activities from space using night-time light

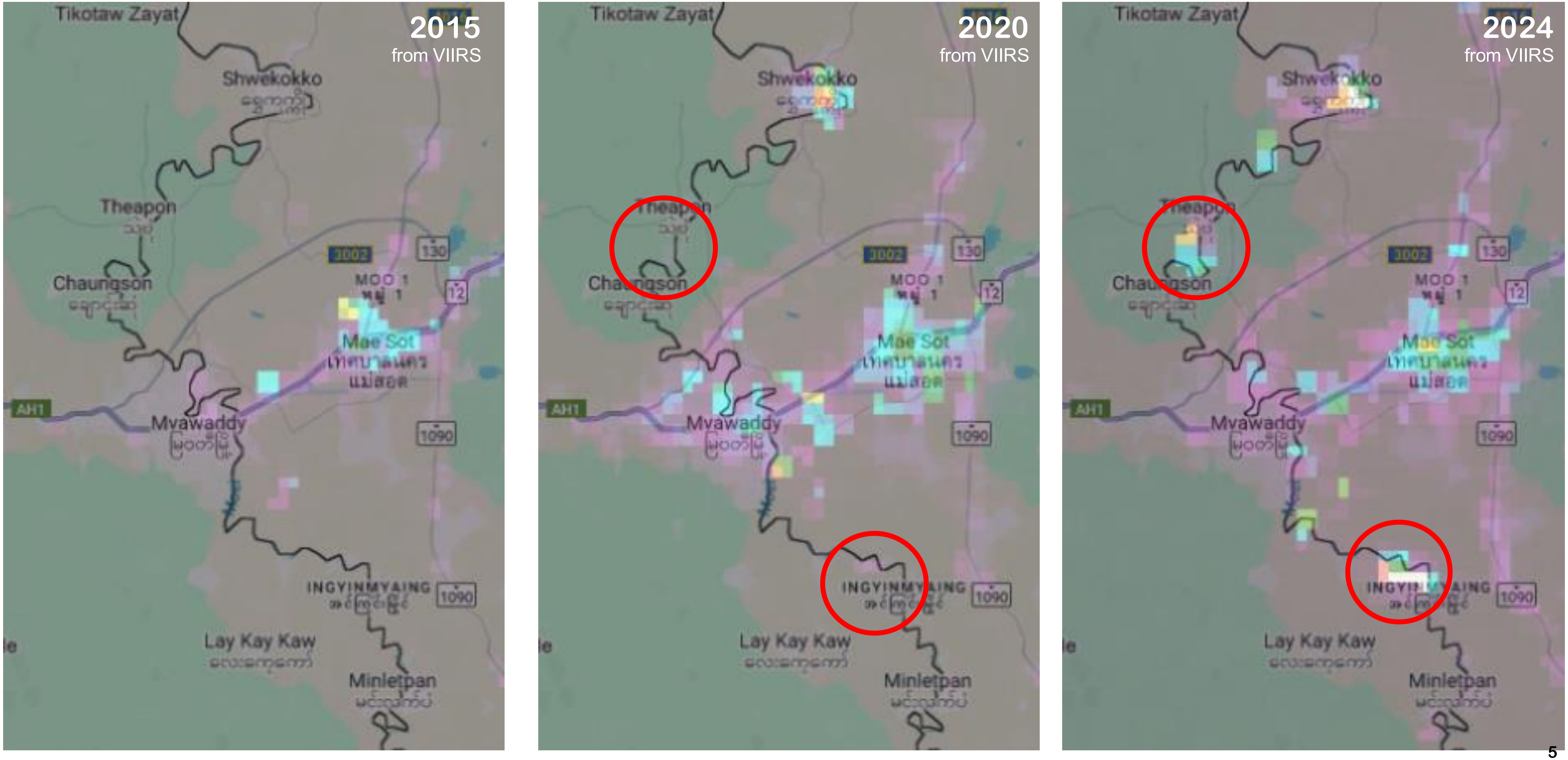
Figure 4: Average night-time light from in 2015 and 2024 from VIIRS calculated by NASA in Shan region



Motivation

Observed activities from space using night-time light

Figure 5: Average night-time light from in 2015, 2020, and 2024 from VIIRS calculated by NASA in Kayin



Research Questions

Q

1) Is there an impact of 2021 coup on economic activities?

2) Informal and illicit economies in border regions significantly expanded after the coup?

Methodology

Table 1: Location of known or reported casinos, scam centres and related as treatment areas

	Region	Name	Name of location	Location	Source
ADM3	Kayin	Myawady	Myawadi	MMR.6.3.1_1	GADM
	Shan	Mongla	Keng Tung	MMR.13.1.1_1	GADM
		Mongpauk	Mong Yang	MMR.13.1.2_1	GADM
		Pangsang	Pang-Yang	MMR.13.4.7_1	GADM
		Laukkaing	Kunlong	MMR.13.2.2_1	GADM
ADM4	Kayin	Myawaddy	Me Htaw Tha Lay	MMR003005001	MIMU
			Me Ka Nei	MMR003005009	MIMU
			Myawaddy	MMR003005701	MIMU
			Shewkokko	MMR003005901	MIMU
	Shan	Mongpauk	Mongyang	MMR016003702	MIMU
		Pangsang	Pangsang	MMR015005701	MIMU
		Mongla	Mongla	MMR016005701	MIMU
		Laukkaing	Tar Shwe Htan	MMR015022014	MIMU
			Htin Par Keng	MMR015022002	MIMU
			Shwe Yin See	MMR015022007	MIMU
			Laukkaing	MMR015022701	MIMU

Note: ADM3 refers to the administrative level 3 used for analysis across Myanmar as a whole. For more detailed analysis in specific regions—namely Kayin and Shan—a finer administrative level (ADM4), provided by MIMU, is used.
Source: UNODC (2024), Global administrative areas (GADM) version 2.8 and Myanmar Information Management Unit (MIMU)

Difference-in-differences

$$NTL_{it} = \alpha + \beta^1 Post_t + \beta^2 Treated_i + \beta^3 (Post_t \times Treated_i) + \delta Controls_{it} + \epsilon_{it}$$

where:

- Y_{it} = Night-time light (DN)
- $Post_t$ = 1 if after the 2021 coup, 0 otherwise
- $Treated_i$ = 1 for TOC area, 0 for other
- β^3 = DiD estimate, capturing the differential impact of the coup
- $Controls_{it}$= vector of control variables

Data Collection

Table 2: Descriptive statistics

Variables	Region	N	Mean	SD	Min	Max
Night-time light (Oct-May: 8M)	Myanmar	3,718	0.325	1.912	0.001	31.713
	Kayin	5,120	0.123	0.362	0.0001	7.803
	Shan	20,934	0.170	1.463	0.0001	133.620
Night-time light (Nov-Apr: 6M)	Myanmar	3,718	0.327	1.939	0.001	32.034
	Kayin	4,913	0.116	0.375	0.0001	7.727
	Shan	20,595	0.173	1.396	0.0001	120.422
Night-time light (Dec-Mar: 4M)	Myanmar	3,718	0.317	1.982	0.0001	32.580
	Kayin	4,280	0.101	0.402	0.0001	8.012
	Shan	19,046	0.165	1.443	0.0001	117.614
ACLED (index)	Myanmar	3,718	19.050	51.417	0	630.5
	Kayin	5,161	0.778	8.905	0	266.0
	Shan	21,138	0.384	5.822	0	511.8
Elevation (Metre)	Myanmar	3,718	432.73	470.26	4.13	2507.34
	Kayin	5,161	233.23	295.31	6.29	1346.51
	Shan	21,138	1024.70	293.89	127.80	1981.50

Note: 1) Night-time light values represent the average from NASA's Black Marble (VNP46A3) during the dry season, shown in digital number units. 2) The ACLED index is calculated similarly to the UNDP's Vulnerability to Conflict Index (VCI), using a weighted average: 75% based on the number of all events (excluding protests) and 25% based on civilian fatalities in each area, as reported by ACLED. 3) Elevation is shown in meters. 4) The "Region" column displays various statistics across Myanmar at the administrative level 3 (ADM3). For Kayin and Shan, a finer administrative level (ADM4) from MIMU is used. 5) For each variable, the table provides the number of observations (N), mean (Mean), standard deviation (SD), minimum, and maximum values.

- Yearly Average NTL during Dry season (Black marble from NASA)
- Myanmar (ADM3)
  Kayin and Shan (ADM4)
- ACLED index created from
  The Armed Conflict Location & Event Data (ACLED)

Parallel trends Assumption

Figure 6: The average NTL of treatment and control location in Myanmar

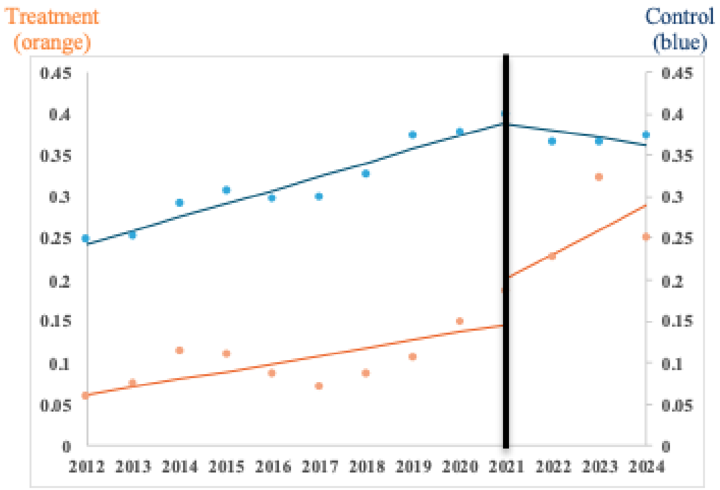


Figure 7: The average NTL in Kayin (Treated locations: Shwe Kokko and Me Htaw Tha Lay)

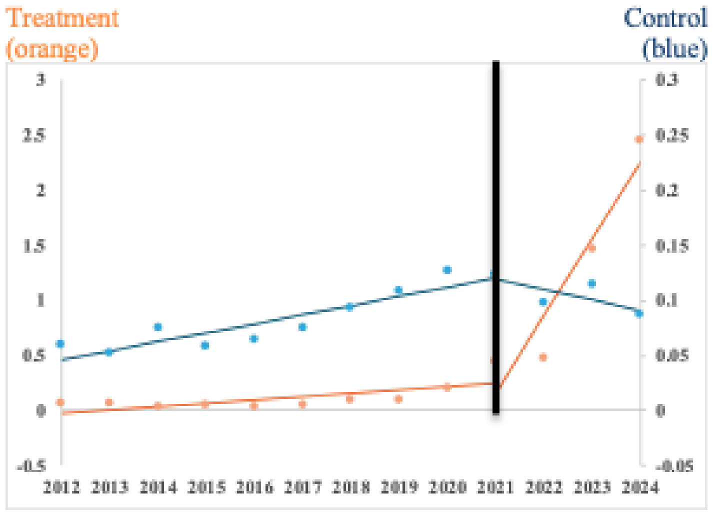
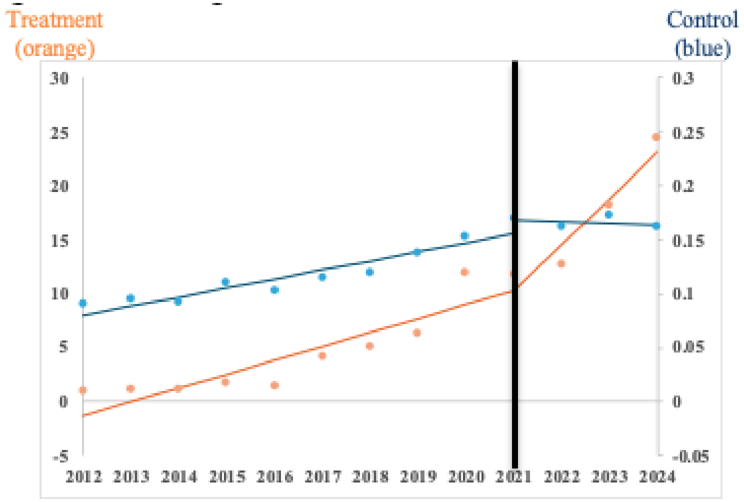


Figure 8: The average NTL of treatment and control location in Shan



Note: ADM3 refers to the administrative level 3 used for analysis across Myanmar as a whole. For more detailed analysis in specific regions—namely Kayin and Shan—a finer administrative level (ADM4), provided by MIMU, is used.
Source: Author's calculation

Event study / Dynamic DiD

Figure 9: Event Study of Difference-in-Differences in Myanmar

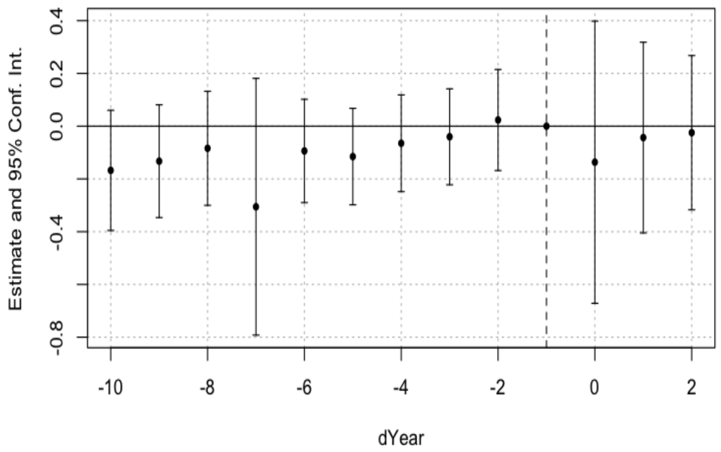


Figure 10: Event Study of Difference-in-Differences in Kayin
(Treated locations: Shwe Kokko and Me Htaw Tha Lay)

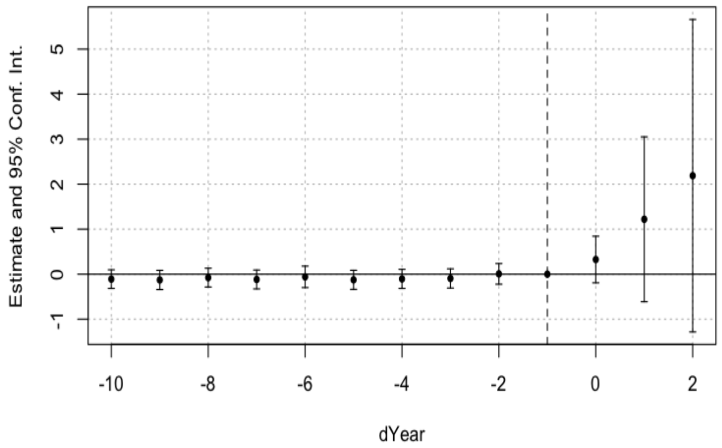
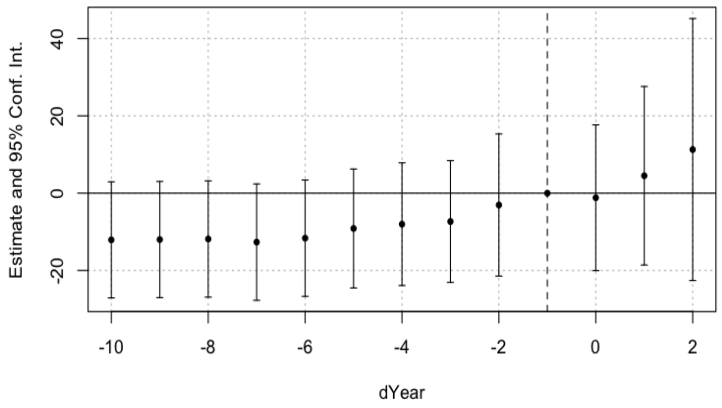


Figure 11: Event Study of Difference-in-Differences in Shan



Note: ADM3 refers to the administrative level 3 used for analysis across Myanmar as a whole. For more detailed analysis in specific regions—namely Kayin and Shan—a finer administrative level (ADM4), provided by MIMU, is used.
Source: Author's calculation

Results

Table 3: The regression results on average night-time light in Myanmar

Y = Average NTL of Myanmar	8M (1.1)	6M (1.2)	4M (1.3)	8M (1.4)	6M (1.5)	4M (1.6)	8M (1.7)	6M (1.8)	4M (1.9)
Treated area	-0.213***	-0.217***	-0.221***	0.054	0.052	0.052	0.051	0.051	0.053
	(0.037)	(0.038)	(0.039)	(0.035)	(0.035)	(0.036)	(0.033)	(0.033)	(0.035)
Post coup d'état	0.051	0.05	0.057	-0.171	-0.176	-0.175	-0.211*	-0.217*	-0.217*
	(0.078)	(0.079)	(0.081)	(0.126)	(0.128)	(0.131)	(0.117)	(0.119)	(0.121)
Treated area*Post coup	0.112	0.101	0.092	0.152	0.142	0.134	0.16	0.149	0.142
	(0.093)	(0.094)	(0.096)	(0.151)	(0.153)	(0.155)	(0.125)	(0.127)	(0.129)
Control variables									
ACLED index				✓	✓	✓	✓	✓	✓
Elevation				✓	✓	✓	✓	✓	✓
Regions							✓	✓	✓
Observations	3718	3718	3718	3718	3718	3718	3718	3718	3718
Adjusted R^2	-0.0005	-0.0010	-0.0005	0.02	0.02	0.02	0.29	0.29	0.29

Note: The analysis covers three types of averaging night-time light during dry season: an 8-month average (October-May), a 6-month average (November-April) and a 4-month average (December-March). Each column presents coefficient estimates with *** p<0.01, ** p<0.05, and * p<0.1, Heteroskedasticity-robust standard errors in parentheses, number of observations, and adjusted R-squared values.

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

1. Myanmar (ADM3)
2. After the coup, Myanmar exhibited an average decline in NTL (−0.217 Digital Number),
3. However, the interaction term for treated areas*post-coup (beta 3) was not statistically significant, indicating no clear evidence of increased illicit activity at this level.

Results

Table 4: The regression results on average night-time light in Kayin region
(Treated locations: Shwe Kokko and Me Htaw Tha Lay)

Y = Average NTL of Kayin	8M (3.1)	6M (3.2)	4M (3.3)	8M (3.4)	6M (3.5)	4M (3.6)	8M (3.7)	6M (3.8)	4M (3.9)
Treated area	0.013 (0.028)	0.019 (0.033)	0.02 (0.035)	-0.006 (0.015)	0.0005 (0.017)	-0.005 (0.017)	-0.189*** (0.062)	-0.189*** (0.065)	-0.162*** (0.062)
Post coup d'état	0.02 (0.013)	0.024* (0.014)	0.023 (0.016)	-0.011 (0.011)	-0.007 (0.011)	-0.018 (0.013)	-0.01 (0.011)	-0.005 (0.011)	-0.018 (0.013)
Treated area*Post coup	1.284** (0.623)	1.293** (0.622)	1.319** (0.640)	1.339** (0.621)	1.348** (0.620)	1.393** (0.638)	1.338** (0.622)	1.346** (0.621)	1.392** (0.639)
Control variables									
ACLED index				✓	✓	✓	✓	✓	✓
Elevation				✓	✓	✓	✓	✓	✓
Provinces							✓	✓	✓
Observations	5120	4913	4280	5120	4913	4280	5120	4913	4280
Adjusted R^2	0.02	0.02	0.02	0.10	0.09	0.11	0.12	0.12	0.14

Note: The analysis covers three types of averaging night-time light during dry season: an 8-month average (October-May), a 6-month average (November-April) and a 4-month average (December-March). Each column presents coefficient estimates with *** p<0.01, ** p<0.05, and * p<0.1, Heteroskedasticity-robust standard errors in parentheses, number of observations, and adjusted R-squared values.
Source: Author's calculation

1. Kayin (ADM4)
2. The interaction term for treated areas post-coup (beta 3) was statistically significant.
3. After the coup treated locations exhibited a statistically significant increase in NTL (+1.3 DN) compared to other areas.

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

Results

Table 5: The regression results on average night-time light in Shan region

Y = Average NTL of Shan	8M (4.1)	6M (4.2)	4M (4.3)	8M (4.4)	6M (4.5)	4M (4.6)	8M (4.7)	6M (4.8)	4M (4.9)
Treated area	4.411***	4.253***	4.267***	4.312***	4.153***	4.165***	6.043***	5.817***	5.929***
	(1.279)	(1.205)	(1.153)	(1.288)	(1.214)	(1.161)	(1.502)	(1.408)	(1.354)
Post coup d'état	0.055***	0.053***	0.047***	0.046***	0.044***	0.039***	0.045***	0.043***	0.038***
	(0.011)	(0.011)	(0.012)	(0.011)	(0.011)	(0.012)	(0.011)	(0.012)	(0.013)
Treated area*Post coup	14.370**	13.630**	13.887**	14.383**	13.643**	13.902**	14.384**	13.644**	13.925**
	(7.161)	(6.630)	(6.588)	(7.162)	(6.630)	(6.588)	(6.868)	(6.341)	(6.285)
Control variables									
ACLED index				✓	✓	✓	✓	✓	✓
Elevation				✓	✓	✓	✓	✓	✓
Provinces							✓	✓	✓
Observations	20934	20595	19046	20934	20595	19046	20934	20595	19046
Adjusted R^2	0.20	0.20	0.21	0.20	0.20	0.21	0.24	0.25	0.25

Note: The analysis covers three types of averaging night-time light during dry season: an 8-month average (October-May), a 6-month average (November-April) and a 4-month average (December-March). Each column presents coefficient estimates with *** p<0.01, ** p<0.05, and * p<0.1, Heteroskedasticity-robust standard errors in parentheses, number of observations, and adjusted R-squared values.
Source: Author's calculation

1. Shan (ADM4)
2. After the coup, Shan exhibited an average increase in NTL (+0.04 Digital Number),
3. The interaction term for treated areas post-coup (beta 3) was statistically significant.
4. After the coup treated locations exhibited a statistically significant increase in NTL (+14 DN) compared to control areas.

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

Conclusion

Q

1) Is there an impact of 2021 coup on economic activities?

2) Informal and illicit economies in border regions significantly expanded after the coup?

1) Yes, at national level, Myanmar exhibited an average decline in NTL (−0.217 DN)

2) Yes, focus on ADM4-level data in high-risk regions.

- Treatment zones in both Kayin and Shan regions experienced significantly greater increases in NTL compared to control areas, indicating heightened human and economic activity, likely associated with the rise of illicit operations.

Policy implications

1. Nighttime Lights (NTL) data presents a valuable opportunity for policymakers to monitor economic activities

2. Spatial lens enables more precise and timely interventions

3. Governments in Southeast Asia should focus on enhancing regional cooperation and investing in data-informed monitoring systems that incorporate remote sensing technologies